# CS 363M — Principles of Machine Learning
# Fall 2026

**Midterm exam · solutions**

Name: _____ EID: _____

Course modality (circle one): In-person Online

## Instructions

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*Read this page before you begin. Do not turn the page until you are told to start.*

- There are **4 questions** worth a total of **70 points**. Point values are shown beside each question and part.
- Budget your time according to the point values. Computational parts generally take longer than conceptual parts.
- **Each question begins on its own page. Write your answer for each part in the blank space directly below it, and label it** with its part number — 1.1, 1.2, and so on.
- **You must submit your cheat sheet with your exam.** Write your name on it and hand it in stapled behind the last page. An exam handed in without its cheat sheet will not be graded until the sheet is produced.
- One double-sided cheat sheet is permitted.
- **Show your work on every computational part.** An answer with no work receives no credit. Correct work with an arithmetic slip receives most of the credit, so write down the steps even when you are unsure of the final number.
- Round to four decimal places unless a part tells you otherwise. Leaving an answer as an unreduced fraction is fine.
- A basic or scientific calculator is permitted. Phones, laptops, tablets and smart watches must be off and out of sight for the whole period.
- If a part seems ambiguous, write down the assumption you are making and answer under that assumption. A clearly stated assumption will be graded fairly.
- Keep your eyes on your own paper. Suspected academic dishonesty is referred to Student Conduct and Academic Integrity.
- If you finish early you may hand in your exam and leave quietly. In the last ten minutes, please stay seated so you do not disturb the others.

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**Good luck.**

## 1. Evaluating classifiers (20 points)

A university wants to use an AI-text detector to check student essays. To test it, the detector is run on 400 essays whose authorship is known: 60 were written with AI and 340 by students. Of the 400 essays, the detector flags 50 as written with AI. Of these 50 flagged essays, 45 really were written with AI. Treat "written with AI" as the positive class.

**1.1 (4 points) Write down the confusion matrix.**

**Solution**

60 AI-written essays and 340 human-written essays.

| | flagged | not flagged |
|---|---|---|
| written with AI | TP $= 45$ | FN $= 15$ |
| written by students | FP $= 5$ | TN $= 335$ |

**1.2 (2 points) What are precision and recall for this detector?**

**Solution**

Precision $= 45/50 = 0.90$. Recall $= 45/60 = 0.75$.

**1.3 (2 points) Do we care more about improving precision or recall for this problem, and why?**

**Solution**

Precision. A false positive accuses an innocent student of cheating; a missed AI essay (false negative) is a smaller harm. A flag should almost always be right before anyone acts on it.

The detector also gives each essay a score between 0 and 1; a higher score means "more likely written with AI". Here are 8 essays, 4 written with AI and 4 by students:

| essay | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| score | 0.95 | 0.85 | 0.70 | 0.60 | 0.50 | 0.40 | 0.30 | 0.10 |
| written with AI? | yes | yes | no | yes | no | yes | no | no |

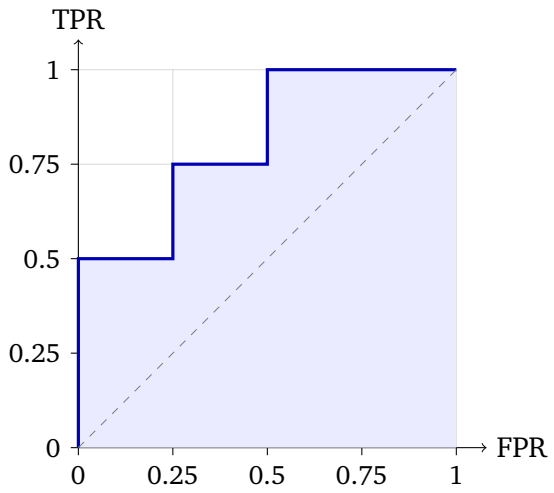
We say the detector flags an essay as AI-written if its score is over $t$.

**1.4 (10 points) Draw the ROC curve.**

**Solution**

As $t$ passes each essay, from the highest score down, an AI essay moves the point up by $1/4 = 0.25$ and a human essay moves it right by $0.25$. Starting at $(0, 0)$, the points are

$$(0, 0.25),\ (0, 0.5),\ (0.25, 0.5),\ (0.25, 0.75),\ (0.5, 0.75),\ (0.5, 1),\ (0.75, 1),\ (1, 1).$$




**1.5 (2 points) Compute the AUC.**

**Solution**

The area under the staircase, in columns of width $0.25$ with heights $0.5, 0.75, 1, 1$: $\text{AUC} = 0.25 \times (0.5 + 0.75 + 1 + 1) = 0.8125 \approx 0.81$.

## 2. Support vector machines (20 points)

**2.1 (10 points)** A soft-margin SVM returned $\boldsymbol{w} = (6, 8)$, $b = -20$.

| point | A | B | C | D | E |
|---|---|---|---|---|---|
| $x_1$ | 3 | 1 | 1.5 | 1.25 | 3 |
| $x_2$ | 1 | 1 | 1.5 | 1.5 | 0.5 |
| label | $+$ | $-$ | $+$ | $-$ | $-$ |

**(a) Compute $f(\boldsymbol{x})$ and the predicted class of each point.**

**(b) What is the width of the margin?**

**(c) Compute each point's distance to the decision boundary.**

**(d) Which points are support vectors? Which point is on the margin, which is inside it, and which is misclassified?**

**Solution**

$f(\boldsymbol{x}) = 6x_1 + 8x_2 - 20$ and $\|\boldsymbol{w}\| = \sqrt{6^2 + 8^2} = 10$.

| point | A | B | C | D | E |
|---|---|---|---|---|---|
| $f(\boldsymbol{x})$ | 6 | $-6$ | 1 | $-0.5$ | 2 |
| predicted class | $+$ | $-$ | $+$ | $-$ | $+$ |
| distance $\lvert f(\boldsymbol{x})\rvert/\|\boldsymbol{w}\|$ | 0.6 | 0.6 | 0.1 | 0.05 | 0.2 |
| $y\, f(\boldsymbol{x})$ | 6 | 6 | 1 | 0.5 | $-2$ |

(b) Margin width $= 2/\|\boldsymbol{w}\| = 2/10 = 0.2$.

(d) The support vectors are the points with $y\, f(\boldsymbol{x}) \le 1$: C, D and E. C is on the margin ($y\, f = 1$), D is inside the margin but correctly classified ($y\, f = 0.5$), and E is misclassified (label $-$, predicted $+$).

**2.2 (4 points)** The soft-margin SVM solves

$$\min_{\boldsymbol{w}, b, \boldsymbol{\xi}} \ \tfrac{1}{2}\|\boldsymbol{w}\|^2 + C \sum_i \xi_i \quad \text{subject to} \quad y_i(\boldsymbol{w} \cdot \boldsymbol{x}_i + b) \ge 1 - \xi_i, \ \ \xi_i \ge 0.$$

**(a) What does the slack $\boldsymbol{\xi_i}$ measure, and why do we need it?**

**(b) Where does a training point lie, relative to the margin and the decision boundary, if $\boldsymbol{\xi_i = 0}$? If $\boldsymbol{0 < \xi_i < 1}$? If $\boldsymbol{\xi_i > 1}$?**

**Solution**

(a) $\xi_i$ measures how far point $i$ falls short of being on its correct side, outside the margin. Without slack, the constraints $y_i(\boldsymbol{w} \cdot \boldsymbol{x}_i + b) \ge 1$ have no solution when the classes overlap; slack lets some points violate the margin, at a cost $C\xi_i$ each.

(b) $\xi_i = 0$: on its correct side, on or outside the margin. $0 < \xi_i < 1$: inside the margin, but still correctly classified. $\xi_i > 1$: on the wrong side of the decision boundary (misclassified). (At the solution $\xi_i = \max(0,\, 1 - y_i f(\boldsymbol{x}_i))$, so these are the cases $y_i f \ge 1$, $0 < y_i f < 1$ and $y_i f < 0$.)

**2.3 (4 points)** Two soft-margin SVMs are trained on the same data, one with $C = 0.01$ and one with $C = 100$.

**(a) Which has the wider margin, and why?**

**(b) Which has more support vectors, and why?**

**(c) Which is more likely to overfit, and why?**

**(d) How would you choose $C$ in practice, and why?**

**Solution**

(a) $C = 0.01$: slack is cheap, so the SVM accepts many points inside the margin in exchange for a wider one. (b) $C = 0.01$: with a wider margin, more points fall on or inside it, and all of them are support vectors. (c) $C = 100$: violations are expensive, so the margin narrows to fit individual training points. (d) $C$ is a hyperparameter: try several values and pick the one with the best accuracy on validation data (or by cross-validation). Training accuracy cannot choose it, because a larger $C$ always fits the training data at least as well; only held-out data shows which $C$ generalizes.

**2.4 (2 points) You remove a training point that is not a support vector and retrain the SVM. What changes? Why?**

**Solution**

Nothing. The solution depends only on the support vectors; a point outside the margin does not affect $\boldsymbol{w}$ or $b$.

## 3. Nonlinear regression (20 points)

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**3.1 (4 points)** kNN regression is trained on six points:

| $x$ | 1 | 2 | 4 | 5 | 7 | 8 |
|---|---|---|---|---|---|---|
| $y$ | 3 | 6 | 4 | 8 | 10 | 12 |

**Predict $y$ at $x = 4.4$ using (a) $k = 1$ and (b) $k = 3$.**

**Solution**

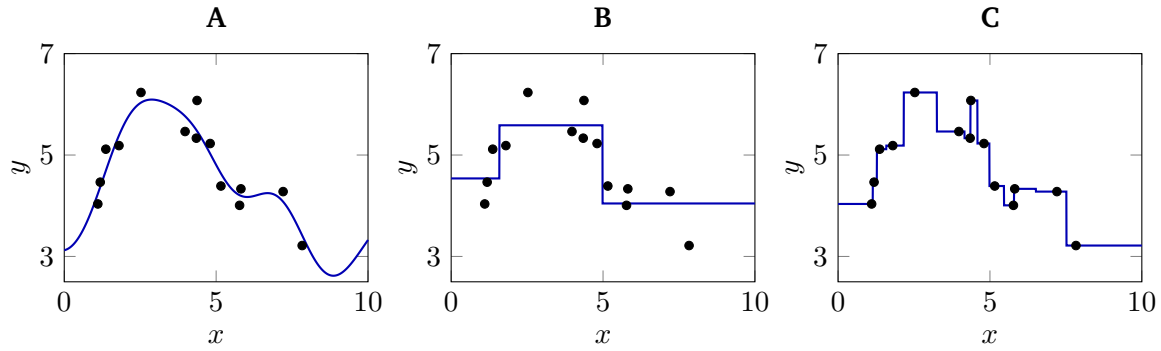
(a) The nearest point is $x = 4$, so the prediction is $4$. (b) The three nearest points are $x = 4, 5, 2$ (distances $0.4, 0.6, 2.4$), so the prediction is $(4 + 8 + 6)/3 = 6$.

**3.2 (6 points) For each of a regression tree, kNN regression and SVR, name a hyperparameter that controls how closely the model follows the training data. Which setting of it makes overfitting more likely?**

**Solution**

Regression tree: a larger maximum depth, or a smaller minimum number of points needed to split. kNN: a smaller $k$. SVR: a larger $C$, or a smaller $\varepsilon$.

**3.3 (5 points) The figure shows three models fitted to the same data. Which panel shows a regression tree, which shows kNN with $k = 1$, and which shows SVR with a kernel? Explain each.**




**Solution**

A is SVR with a kernel: a smooth curve. B is the regression tree: a few wide flat steps, one for each leaf, which predicts the average $y$ of its points. C is kNN with $k = 1$: a narrow step for every training point, passing through every point.

**3.4 (5 points) We use kNN to predict house prices from floor area (500 to 5,000 square feet) and number of bedrooms (1 to 6). Why might kNN do badly on these raw features? What would you do?**

**Solution**

Distances are dominated by floor area: a difference of one bedroom counts as much as one square foot, so the nearest houses match on area and almost ignore bedrooms. Standardize each feature (subtract its mean, divide by its standard deviation) before computing distances.

## 5. Ensemble methods (10 points)

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**5.1 (5 points)** An SVM can only separate two classes. A clinic wants to classify chest X-rays into 4 classes: normal, pneumonia, COVID and tuberculosis.

**(a) With one-vs-rest partitions, how many SVMs do you train, and what labels does each one see?**

**(b) How is a new X-ray classified?**

**(c) With one-vs-one partitions, how many SVMs do you train?**

**Solution**

(a) 4 SVMs, one per class. Each sees "this class" against "the other three". (b) Each SVM predicts its side, and every class on the predicted side gets one vote; the class with the most votes wins. (Also accept: pick the class whose SVM gives the largest $f(\boldsymbol{x})$.) (c) One SVM for every pair of classes: $4 \cdot 3/2 = 6$.

**5.2 (3 points) An ensemble of 25 classifiers is built, but each base classifier has error rate 0.6. What happens to the error of the majority vote, compared with a single classifier? Why?**

**Solution**

The ensemble is worse than a single classifier: with independent errors, the majority vote is wrong whenever more than half of the classifiers are wrong, and with error 0.6 that happens most of the time (about 0.85). A majority vote only helps when each base classifier is better than random guessing.

**5.3 (2 points) Random forests consider only a random subset of the features at each split. Why does this help the ensemble?**

**Solution**

It makes the trees less alike. Without it, every tree would split on the same strong features and make similar mistakes; with random subsets the trees are more diverse, so their errors are less correlated and the vote cancels more of them.